

APPENDIX K – BEST-OPTIMIZED ENSEMBLE METHODS FOR CODE SMELL DETECTION

EM	Feature Selection (FS)	Data Scaling for FS	% Predictors for FS	Target for FS	Hyperparameter Optimization	Acc	Mean	Std	Time(s)
XGB	Chi-square	Normalization	25%	Binary	Randomized	95.56	92.98	1.75	448
RF	Chi-square*	Normalization	30%	Binary	Randomized	95.16	92.73	1.39	33
XGB	Chi-square	Normalization	30%	Binary	Bayesian	95.16	92.73	2.37	3551
RF	Chi-square	Normalization	25%	Binary	Randomized	94.76	92.61	1.41	35
CB	Chi-square*	Normalization	30%	Binary	Randomized	94.35	92.49	2.05	1697
XGB			all features		Bayesian	94.35	93.58	0.98	8169
CB			all features		Bayesian	94.35	93.58	1.98	8407
RF			all features		Randomized	93.95	92.37	2.19	53
XGB	Anova	-	30%	Multiclass	Randomized	93.95	93.83	1.77	379
CB	Anova	-	30%	Multiclass	Randomized	92.74	93.58	1.13	3395
RF	Chi-square	Normalization	20%	Multiclass	Bayesian	94.76	93.58	1.26	123
CB	Chi-square	Normalization	20%	Multiclass	Bayesian	93.55	93.58	1.26	3239
CB	Anova	-	25%	Binary	Bayesian	93.55	93.58	1.56	2511
DT	Anova	-	15%	Binary	Randomized	91.13	87.04	2.87	1
DT	Chi-square*	Normalization	30%	Binary	Randomized	90.73	89.22	1.57	3
DT	Chi-square	Normalization	30%	Multiclass	Bayesian	90.73	86.20	2.78	346

*Chi-square used by Santos, Duarte e Choren(3) based on Dewangan et al.(4)